Exercise

Consider a sliding puzzle consisting of a 3x3 field with 8 tiles and an empty field. The tiles are labeled with the numbers 1 – 9 and can be pushed vertically or horizontally into the adjacent empty space. The aim of the game is to get the tiles in the following arrangement:

1	2	3
4	5	6
7	8	

The following **starting arrangement is given**:

1		3
4	2	6
7	5	8

Create the resulting A* search tree based on the starting arrangement above to find the moves that transfer the start to the target configuration:
- Number the resulting nodes so that the order of expansion is visible
- You can represent the empty space with a 0

Use the following evaluation function:
- for the costs from the start: number of moves
- as a heuristic function: distance sum of the tiles of the tiles

The distance sum of the tiles is the sum of the distances of all tiles from their target position, whereby the gap is not summed up. The distance of a single tile from its target position is calculated as the sum of the distances in the horizontal and vertical directions.

Example configuration

1	5	7
2	8	4
3		6

Distances of the tiles

0	1	4
2	1	2
4		1

Distance sum: 15